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BUILDING MATERIAL AND CONSTRUCTION **Course Code : 312338**

Programme Name/s	**: Civil Engineering/ Civil & Rural Engineering/ Construction Technology/ Civil & Environmental Engineering/**
Programme Code	**: CE/ CR/ CS/ LE**
Semester	**: Second**
Course Title	**: BUILDING MATERIAL AND CONSTRUCTION**
Course Code	**: 312338**

I. RATIONALE

Building Materials and Construction is the key element in the construction project. It is a challenging job for the civil engineer to select relevant material for construction which is durable, economical and eco-friendly along with the construction procedure. At diploma level, students are expected to develop their understanding, performance-oriented abilities in order to apply their knowledge in construction industry. This course essentially imparts the knowledge of construction technology along with the processes involved in it and various construction materials used for economic and effective execution of various construction activities. This knowledge shall be used for effective and efficient utilization of these materials during the building construction.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Undertake safe building construction practices with relevant building materials.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Identify relevant type of construction materials for the given type of building.
- CO2 - Use the relevant type of special purpose construction materials in the given situation.
- CO3 - Undertake the given type of building construction activity for the given component of building structure.
- CO4 - Design the relevant means of communication for the given building structure.
- CO5 - Use the relevant type of material for finishing purpose in the given situation.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme												
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical							
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA			
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min		
312338	BUILDING MATERIAL AND CONSTRUCTION	BMC	DSC	3	-	2	3	8	4	3	30	70	100	40	25	10	-	-	25	10	150	

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Total IKS Hrs for Sem. : 1 Hrs
Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment
Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Classify the given type of material used in the given building structure TLO 1.2 Classify the given construction material according to its sources with examples. TLO 1.3 Propose the relevant natural construction material for the given situation. TLO 1.4 Suggest the relevant type of artificial material for the given type of construction work TLO 1.5 Classify the buildings using NBC guidelines	**Unit - I Overview of construction Materials** 1.1 Scope of construction materials in various Civil Engineering Sectors. 1.2 Broad classification of materials – Sources of materials, Natural, Artificial- special, finishing and recycled. 1.3 Natural Building construction Materials – Stone, Timber, Soil, Sand and Coarse Aggregates, Bitumen: Types and uses. (IKS-Materials used in Ancient Buildings-Stone , Lime) 1.4 Artificial Building Construction Materials – Cement, Clay Brick, Flooring Tiles, Concrete Blocks, Plywood, particle board, Veneers, laminated board and Glass: Types and uses. 1.5 Introduction to National Building Code-Part III (2005) Group A to I As per Types of Constructions- Load Bearing Structures, Framed Structures,Composite Structures.	Chalk-Board Demonstration Video Demonstrations Presentations Site/Industry Visit

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Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Describe the method used for water proofing in the given situation. TLO 2.2 Justify the use of fibers in given situation. TLO 2.3 Enumerate the importance of geopolymer cement in construction.	**Unit - II Special Purpose Building Construction Materials** 2.1 Special Building Construction Materials – Waterproofing, Termite proofing, Thermal and sound insulating: Types and suitability. 2.2 Fibers– Jute, Glass, Plastic Asbestos Fibers: Types and uses 2.3 Geopolymer cement: Geo-cement: properties and applications.	Chalk-Board Demonstration Video Demonstrations Site/Industry Visit Presentations Case Study
3	TLO 3.1 Explain the roles and functions of given building components in civil structure TLO 3.2 Describe the process of earthwork excavation for given construction activity. TLO 3.3 Suggest relevant materials used for formwork in the given situation. TLO 3.4 Justify the type of foundation proposed in the given situation with its salient features. TLO 3.5 Undertake the construction of stone masonry in given situation. TLO 3.6 Undertake the construction of Brick masonry in given situation. TLO 3.7 Justify the necessity of scaffolding in construction.	**Unit - III Construction of substructure & Superstructure** 3.1 Building Components: Building Components & their Function: Substructure, Superstructure 3.2 Earthwork: Excavation For Foundation, Timbering and Strutting Earthwork for Embankment Material for Plinth Filling 3.3 Formwork: Definition, Requirements, Materials used, Types and Removal of Formwork. 3.4 Foundation: Functions, Types :Shallow Foundation-Stepped Footing, Wall Footing, Column Footing, Isolated and Combined Column Footing, Raft Foundation. Deep Foundation-Pile Foundation, Well foundation and Caissons, Pumping Methods of Dewatering, Deep wells, Cofferdams. 3.5 Stone Masonry: Terms used in stone masonry- facing, backing, hearting, through stone, corner stone, cornice. Type of stone masonry: Rubble masonry, Ashlar Masonry and their types. Selection of Stone Masonry. Precautions to be observed in Stone Masonry Construction. (IKS- Ancient heritage building-stone masonry work) 3.6 Brick masonry: Terms used in brick masonry- header, stretcher, closer, quoins, course, face, back, hearting, bat bond, joints, lap, frog, line, level and plumb. Bonds in brick masonry-header bond, stretcher bond, English bond and Flemish bond. Requirements of good brick masonry. Precautions to be observed in Brick Masonry Construction ,Comparison between stone masonry and Brick Masonry, Tools and plants required for construction of stone masonry and brick masonry. 3.7 Scaffolding , Shoring and Underpinning: Necessity, types, application. Process of Erection and Dismantling.	Chalk-Board Site/Industry Visit Model Demonstration Video Demonstrations Case Study Presentations Site/Industry Visit

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Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Classify the given types of doors based on its location, material used and dimension. TLO 4.2 Classify the relevant types of windows based on location, material and dimension. TLO 4.3 Select the relevant type of fixtures with fastener for fixing the given type of door or window. TLO 4.4 Classify the staircase on the basis of its shape and material use. TLO 4.5 Suggest the type of staircase for the given situation.	**Unit - IV Building Communication** 4.1 Horizontal Communication: Doors –Components of Doors, Types of Doors: Fully Paneled Doors, Partly Paneled and Glazed Doors, Flush Doors, Collapsible Doors, Rolling Shutters, Revolving Doors, Glazed Doors. Sizes of Door recommended by BIS. 4.2 Windows: Component of windows, Types of Windows: Fully Paneled, Partly Paneled and Glazed, Wooden, Steel, Aluminum Windows, Sliding Windows .Sizes of Windows recommended by BIS and Ventilators 4.3 Fixtures and fastenings for doors and windows. 4.4 Vertical Communication - Stair Case, Ramps, Lift, Elevator and Escalators. Terms used in staircase, Types of staircases- Straight, doglegged, open well, Circular, Quarter turn. Calculation of no of flight/s, dimensions of rise and trade.	Model Demonstration Chalk-Board Video Demonstrations Site/Industry Visit Presentations
5	TLO 5.1 Suggest relevant type of flooring material for for given situation. TLO 5.2 Explain the procedure for laying and Construction of floor. TLO 5.3 Describe the Procedure of Plastering of given thickness. TLO 5.4 Select the relevant type of paint for the given surface area of the building.	**Unit - V Building Finishes** 5.1 Types of Floor Finishes, laying process and its suitability- Shahabad, Kota, Marble, Granite, Kadappa, Ceramic Tiles, Vitrified, Pavement Blocks, Concrete Floors, wooden Flooring, Skirting And Dado. 5.2 Plastering – Necessity, Procedure, Single Coat and Double Coat Plaster, rough finish, Neeru Finishing and POP. 5.3 Special Plasters- Stucco Plaster, sponge finish, pebble finish. Plaster Board And Wall Claddings. 5.4 Painting –Necessity, Surface Preparation for painting, Methods of Application, Selecting Suitable Painting Material.	Site/Industry Visit Video Demonstrations Presentations Demonstration Chalk-Board

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Identify the different Construction materials used in a construction	1	*Identify minimum three available construction materials in the laboratory and prepare a report with photos/pictures/sketches including writeup on its sources and utility.	2	CO1

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 2.1 Identify the grain distribution pattern used in a construction	2	Identify the grain distribution pattern of the given sample of wood material available in the laboratory and draw the various patterns to prepare concise report on it. (along and perpendicular to the grains)	2	CO1
LLO 3.1 Identify various layers and types of soil strata in foundation pit	3	Prepare the inspection report with relevant photographs by inspecting the three pits of foundation of a site to Identify the different types of layers of soil strata	2	CO1
LLO 4.1 Record dimensions of given bricks	4	*Record the dimensions of 10 bricks to find its average dimension, weight with relevant interpretation report.	2	CO1
LLO 5.1 Perform field test on given sample of brick	5	*Perform field tests on given sample of brick such as- dropping, striking and scratching by nail and interpret the results obtained to decide its quality and prepare a report on it.	2	CO2
LLO 6.1 Apply the relevant termite chemical to prevent the surface damage	6	Apply the relevant termite chemical on given damaged surface of timber and submit the observation report after one month with photos/pictures.	2	CO2
LLO 7.1 Paint the given surface of wall after preparing a required base of relevant material	7	Apply two or more coats of selected paint on the prepared base of a given wall surface for the area of 2m x 2m using relevant tools brush/rollers adopting safe practices and prepare a report on it.	2	CO2
LLO 8.1 Prepare the cement mortar of given proportion	8	Prepare the cement mortar of proportion 1:3 or 1:6 using artificial sand as a special processed construction material and prepare a report on it with sketches/photos while preparation of mortar.	2	CO3
LLO 9.1 Assemble one and half Brick thick wall in given bond.	9	*Assemble one and half Brick thick wall in a English Bond and prepare a report on it with pictures/photos.	2	CO3
LLO 10.1 Assemble one and half Brick thick wall in given type of bond.	10	Assemble Brick thick wall in a Flemish Bond. (minimum 3 Course) and prepare a report on it with sketches/photos.	2	CO3
LLO 11.1 Prepare a site visit report with reference to following: stone masonry, construction site, components of staircase, components of doors & windows, types of flooring, process of plastering, pointing	11	Prepare a visit report with sketches/photos by arranging visit to stone masonry construction work.	2	CO3
LLO 12.1 Prepare a site visit report with reference to following: stone masonry, construction site, components of staircase, components of doors & windows, types of flooring, process of plastering, pointing	12	Prepare a visit report with sketches/photos of construction site with respect scaffolding, formwork and centering work.	2	CO3

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 13.1 Prepare a site visit report with reference to following: stone masonry, construction site, components of staircase, components of doors & windows, types of flooring, process of plastering, pointing	13	*Prepare report with labelled sketches of inspected staircase components during site visit.	2	CO4
LLO 14.1 Prepare a site visit report with reference to following: stone masonry, construction site, components of staircase, components of doors & windows, types of flooring, process of plastering, pointing	14	*Prepare report with labelled sketches of inspected doors and windows components during site visit.	2	CO4
LLO 15.1 Prepare a site visit report with reference to following: stone masonry, construction site, components of staircase, components of doors & windows, types of flooring, process of plastering, pointing	15	Prepare report with labelled sketches of inspected flooring and roofing materials during site visit.	2	CO5
LLO 16.1 Prepare a site visit report with reference to following: stone masonry, construction site, components of staircase, components of doors & windows, types of flooring, process of plastering, pointing	16	*Prepare a visit report with sketches/photos by observing the process of plastering and pointing of a masonry work at construction site.	2	CO5
LLO 17.1 Prepare a site visit report with reference to following: stone masonry, construction site, components of staircase, components of doors & windows, types of flooring, process of plastering, pointing	17	Prepare a visit report with sketches/photos by observing keenly the process of painting in residential / public building.	2	CO5
LLO 18.1 Carry out market survey of construction materials	18	*Carry out market survey of the building materials used for Brickwork,Flooring,Plastering and Painting, available in your city & prepare a report (each of five).	2	CO1 CO2
LLO 19.1 Prepare the site visit report of the nearby heritage structure	19	Prepare the site visit report of the nearby heritage structure to inspect the Civil Engineering attributes with reference to IKS.	2	CO1 CO3

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Micro project

- Collection of information related to different techniques of demolition of existing structure.
- Collect the market rates for following construction materials from various dealers/suppliers of local market for different brands. i. Bricks. ii. Stone / aggregate (20 mm, 40 mm and 80 mm) iii. Teak wood. iv. Flooring tiles. v. Ordinary Portland Cement vi. Oil paint vii. Cement Paint viii. Plaster of Paris ix. Plastic paints x. Recent types of paint.
- Collect the technical brochures of following construction materials. i. Ordinary Portland Cement ii. Vitrified flooring tiles. iii. Particle boards used for aluminum partitions. iv. Paints.
- Undertake a market survey for the cost and technical specification of different brands of following construction Materials and prepare comparison chart. i. Cement ii. Tiles iii. Glass iv. Paints.
- Collection of information related to recent technologies used in building construction.
- Identify the different types of cracks and remedial measures for existing structure (Case Study).
- Visit to the site to check different construction activities as per the check list.

Assignment

- Other than the classroom and laboratory learning, following are the suggested student-related co-curricular activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports.
- Undertake a market survey of different construction materials and compare the following points. i. Structure ii. Properties iii. Applications.
- Prepare journals consisting of sketches of construction materials.
- Undertake a market survey from local dealers for procurement of civil engineering material.
- Inspect the various activities related to Construction material at sites of different civil structures.
- Literature survey of available at institute library regarding construction material used for different purposes and situations.
- Develop Power point presentation or animation for demonstrating laying and fixing the construction materials.
- Classify the buildings with reference to National Building Code- Part III (2005). ii. Identify the components of a building by observing the model. iii. Organize the visit to construction site to observe brickwork, Sill, Lintel, Chajja, Slab, Parapet wall, flooring.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

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Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Saw of different types (Rip saw having 4 to 6 mm pitch, cross cut saw with tooth pitch 2 to 3 mm, panel saw)	2
2	Bricks and blocks of different sizes.	4
3	Steel Tape	4,7,13,14,15
4	Weighing balance	4,7,8
5	Pan, spade	4,7,8
6	Painting brushes of different size for oil, acrylic painting and rollers of different size for smooth finishing work.	7
7	Paints-OBD, acrylic, plastic emulsion.	7
8	Trowels (Brick, Buttering, Pointing) , triangular, ranging in size up to about 11 inches (279.40 mm) long and from 101.6 mm to 203.2 mm wide i.e. (4 to 8 inches wide).	7,8
9	Ordinary Portland Cement, PPC	8
10	Portable Hammer,Spade, Pans (Ghamela), Thread, lime	9,10
11	Square, mason’s level, and straightedge 28.57 mm to 38.10 mm and the middle portion of the top edge from 152.40 mm to 254 mm wide	9,10
12	Models: a) Cut section of building showing different components b) Types of Bonds in Brick masonry c) Types of Door and Windows d) Types of Stairs e) Types of Roofs f) Formwork for different RCC elements g) Types of scaffolding	9,10,13

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Overview of construction Materials	CO1	7	4	4	4	12
2	II	Special Purpose Building Construction Materials	CO2	6	0	4	4	8
3	III	Construction of substructure & Superstructure	CO3	14	4	12	8	24
4	IV	Building Communication	CO4	12	2	6	8	16
5	V	Building Finishes	CO5	6	0	6	4	10
Grand Total				**45**	**10**	**32**	**28**	**70**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Term work, Self-Learning (Assignment) , Question Answer in Classroom, Quiz and Group Discussion. Each practical will be assessed considering- 60% weightage to process and 40% weightage to product.

Summative Assessment (Assessment of Learning)

- Pen Paper test / Oral Exam/ Practical Exam

XI. SUGGESTED COS - POS MATRIX FORM

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Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	2	1	-	1	1	1	1			
CO2	2	1	-	1	2	1	1			
CO3	3	2	1	2	2	1	2			
CO4	3	2	1	2	2	1	2			
CO5	3	2	1	2	1	1	2			

Legends :- High:03, Medium:02,Low:01, No Mapping: -
*PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Ghose, D. N.	Construction Materials	Tata McGraw Hill, New Delhi, 2014 ISBN: 9780074516478
2	Rangwala, S.C.	Engineering Materials	Charator publisher, Ahemdabad, 2015, ISBN: 9789385039171
3	S. P. Arora and Bindra	Building Construction	Dhanpat Rai Publication, Delhi Edition 2013,ISBN: 9788189928803
4	S. C. Rangawala	Building Construction	Charotar Publication,Dist-Anand ISBN-10: 8185594856 ISBN-13: 978-8185594859
5	Sushil Kumar	Building Construction	Standard Publication, Edition 2010,ISBN: 9788180141683, 8180141683
6	BIS	National Building Code	Bureau of Indian Standard, New Delhi
7	BIS	BIS 962-1989 Code of Architectural and Building Drawing	Bureau of Indian Standard, New Delhi
8	BIS	BIS 1038- 1983 Steel Doors, Windows and Ventilators	Bureau of Indian Standard, New Delhi

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://www.youtube.com/watch?v=XsFeVuVQE-E	Introduction to Building Materials
2	https://www.youtube.com/watch?v=C6x_ersOn_o	Building Blocks of Bharat
3	https://www.youtube.com/watch?v=3XGt-p-hpdU	Brick Masonry Construction
4	https://www.youtube.com/watch?v=L-VGe2j53NU	15 Essential Tips for Building a 4" Thick Brick Masonry Wall: Expert Construction Guide

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Sr.No	Link / Portal	Description
5	https://www.youtube.com/watch?v=Yg4BLy7f-iI	Introduction to fix formwork for column at site
6	https://www.youtube.com/watch?v=fDKRtQqKzJM	Steps of Plastering

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

MSBTE Approval Dt. 01/10/2024 **Semester - 2, K Scheme**